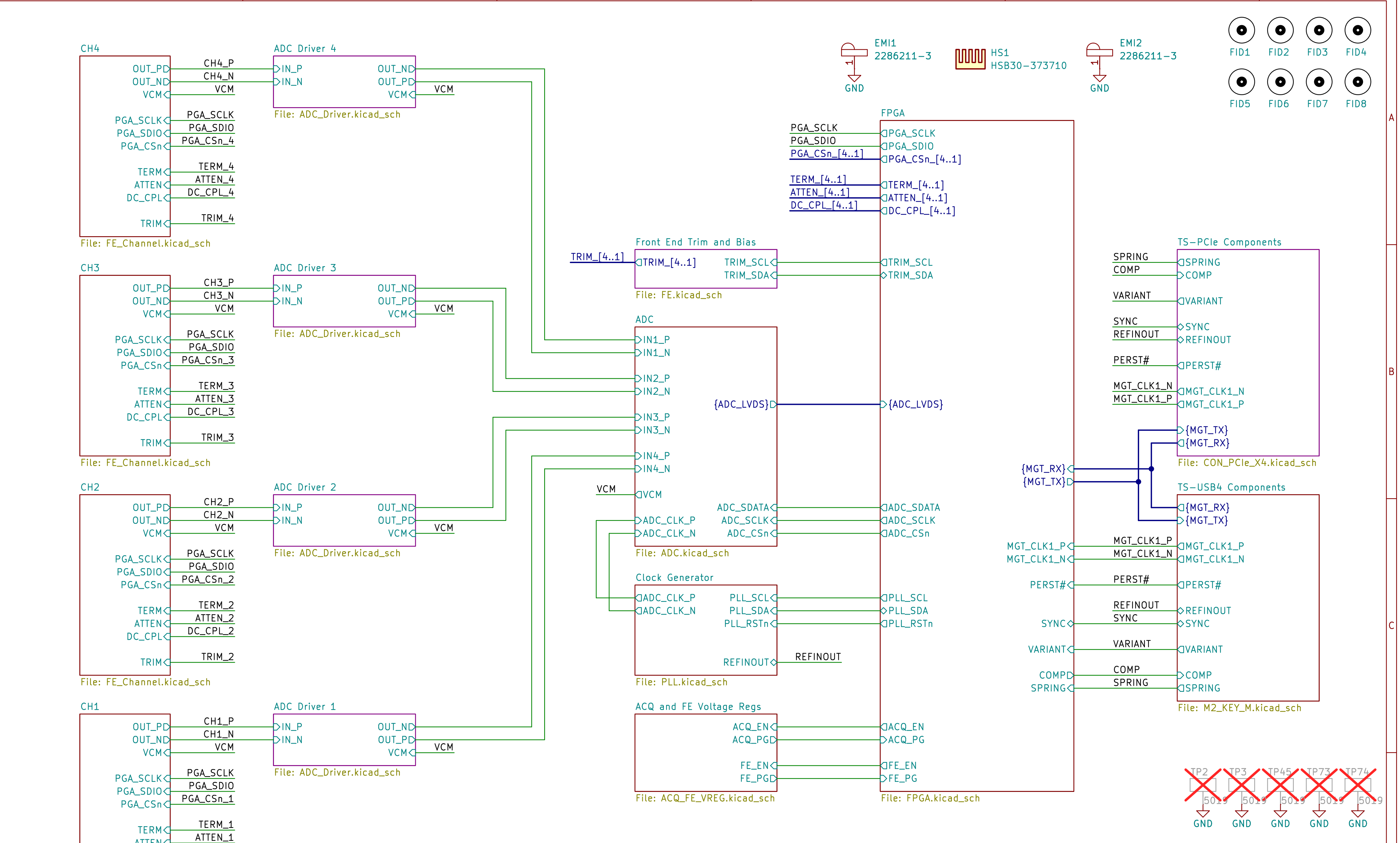
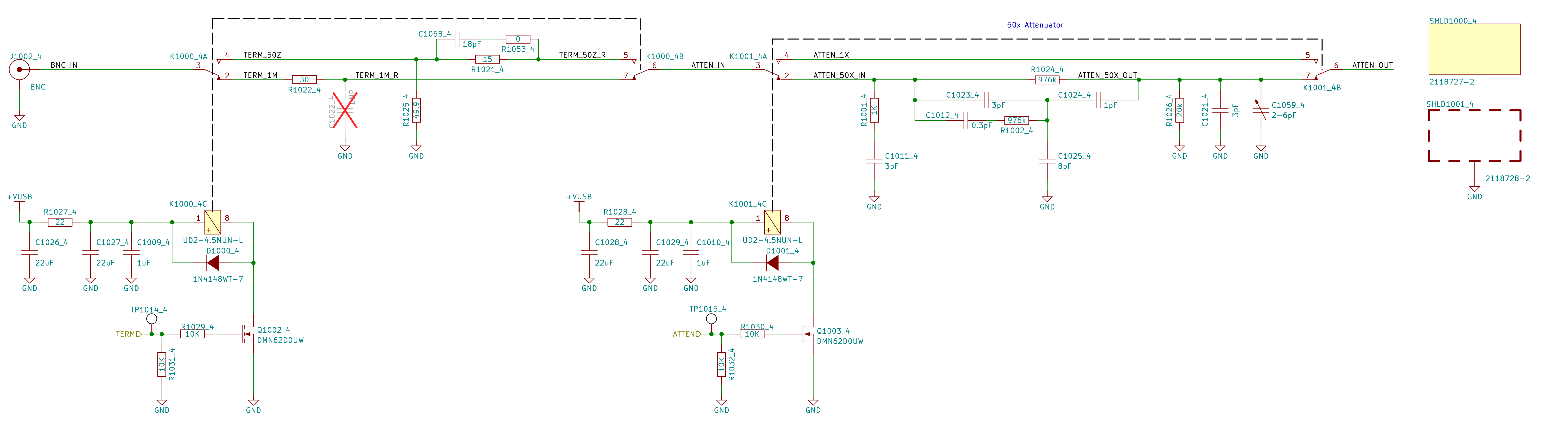


EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /	
File: Thunderscope_Rev5.3.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 1/20

Termination and Attenuation



Input Buffer and AC/DC Coupling

Max current for 35.7kOhm R_Bias: 24mA @ V+ and 24mA @ V-
100k R_Bias wil be lower: 17mA per rail was measured on the EVM

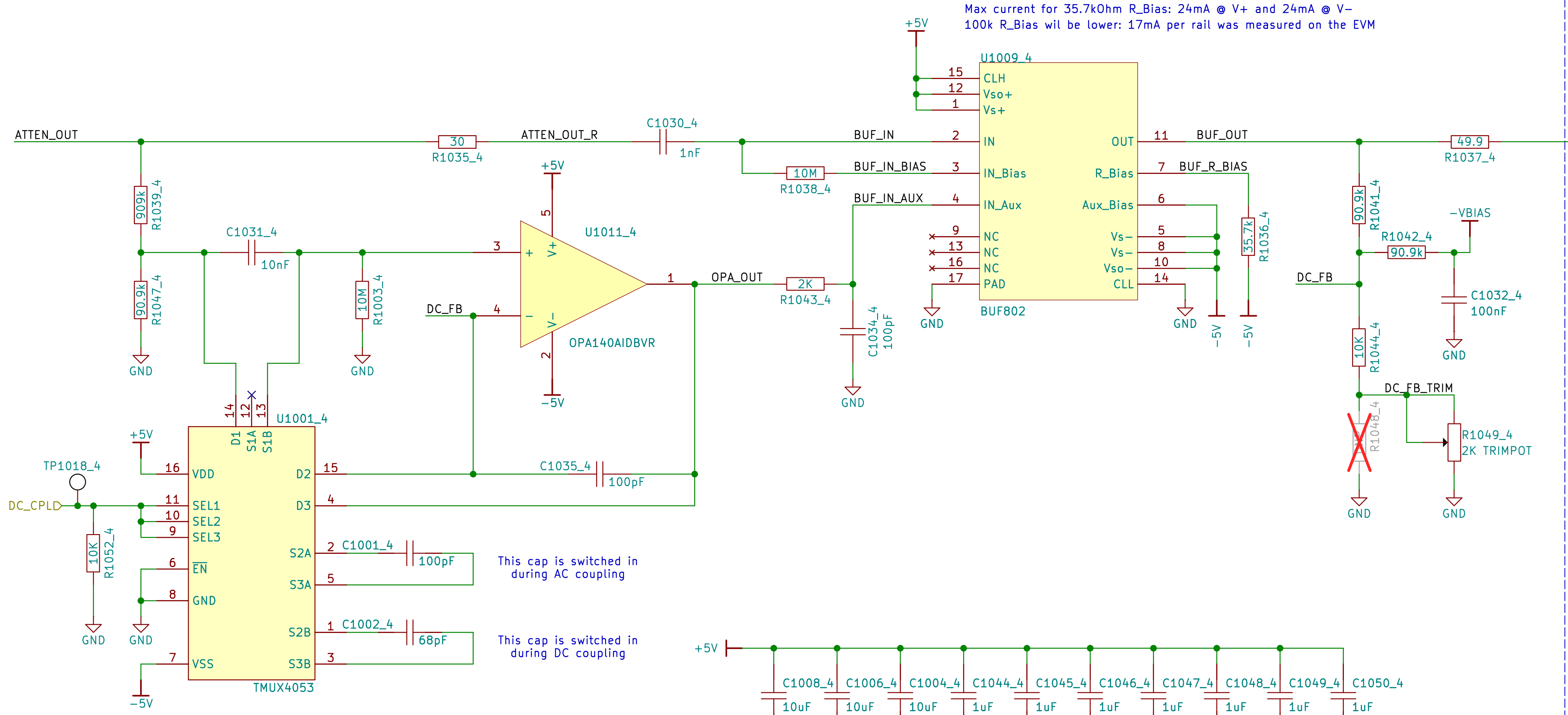
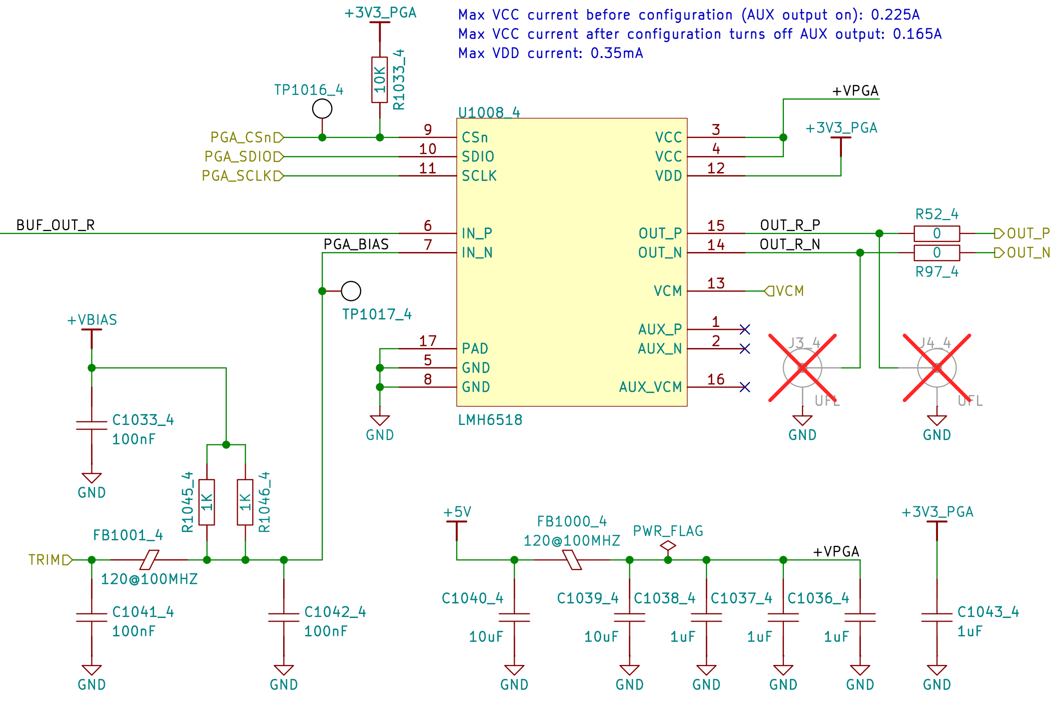


Table 8-3. TMUX4053 Truth Table

$\overline{EN}$	SEL1	SEL2	SEL3	Selected Signal Path Connected To Drain Pins
0	0	X	X	S1A to D1
0	1	X	X	S1B to D1
0	X	0	X	S2A to D2
0	X	1	X	S2B to D2
0	X	X	0	S3A to D3
0	X	X	1	S3B to D3
1	X(1)	X(1)	X(1)	All inputs are unselected (HI-Z)

Programmable Gain Amplifier

Max VCC current before configuration (AUX output on): 0.225A
Max VCC current after configuration turns off AUX output: 0.165A
Max VDD current: 0.35mA



EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /CH4/	
File: FE_Channel.kicad_sch	
Title: ThunderScope	
Size: A3 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 2/20

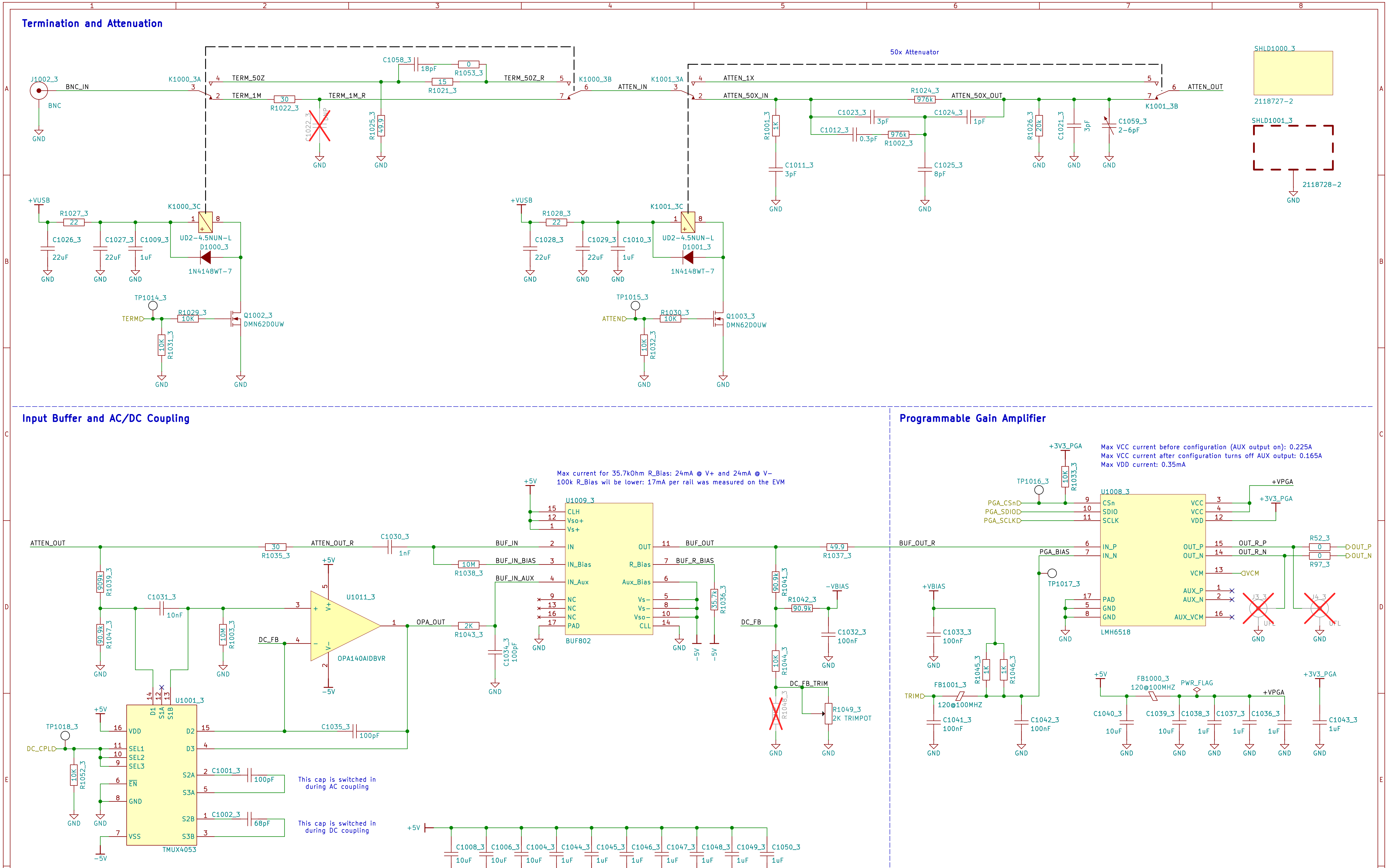


Table 8-3. TMUX4053 Truth Table

EN	SEL1	SEL2	SEL3	Selected Signal Path Connected To Drain Pins
0	0	X	X	S1A to D1
0	1	X	X	S1B to D1
0	X	0	X	S2A to D2
0	X	1	X	S2B to D2
0	X	X	0	S3A to D3
0	X	X	1	S3B to D3
1	X(1)	X(1)	X(1)	All inputs are unselected (HI-Z)

EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /CH3/	
File: FE_Channel.kicad_sch	
Title: ThunderScope	
Size: A3 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 2/20

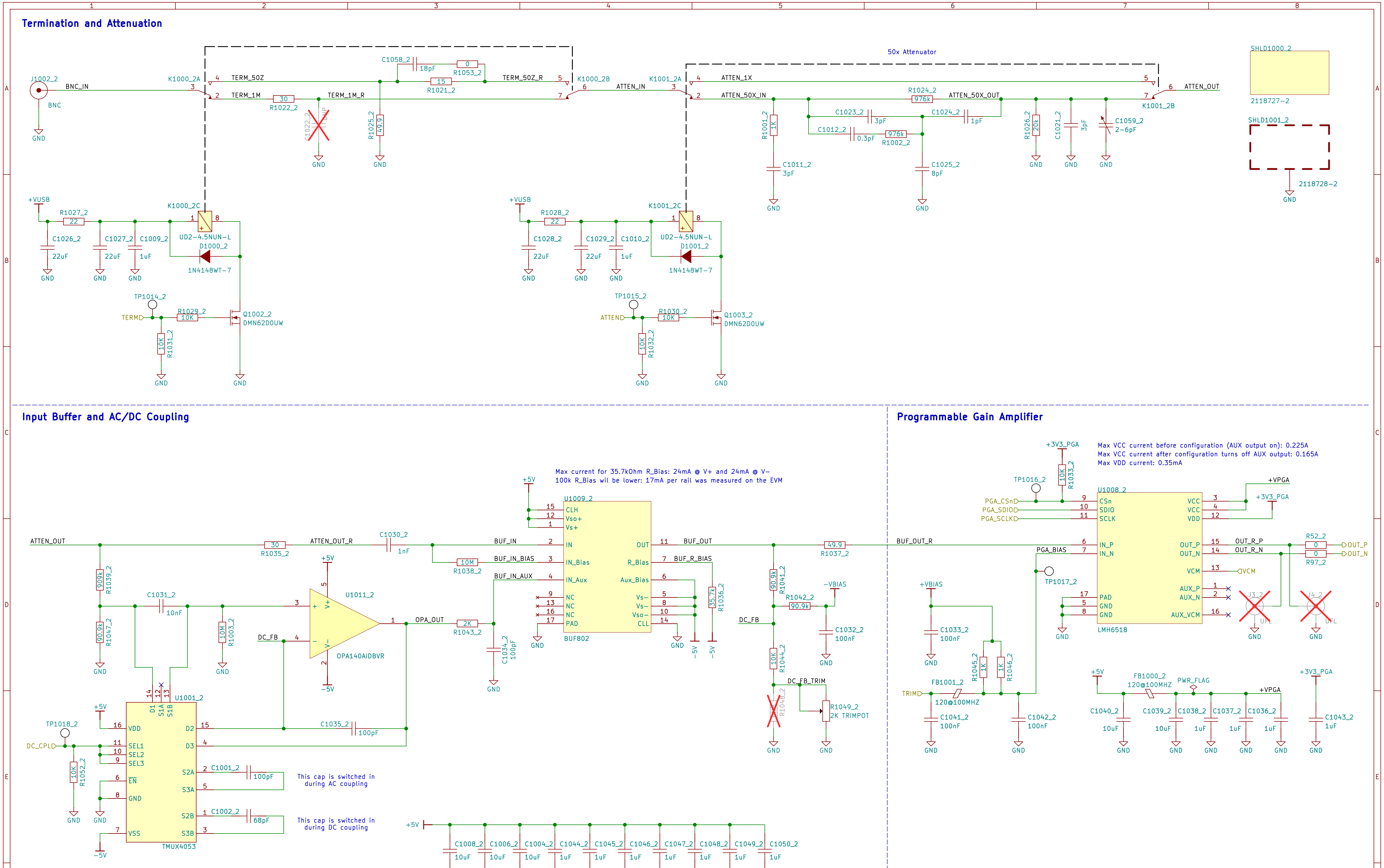
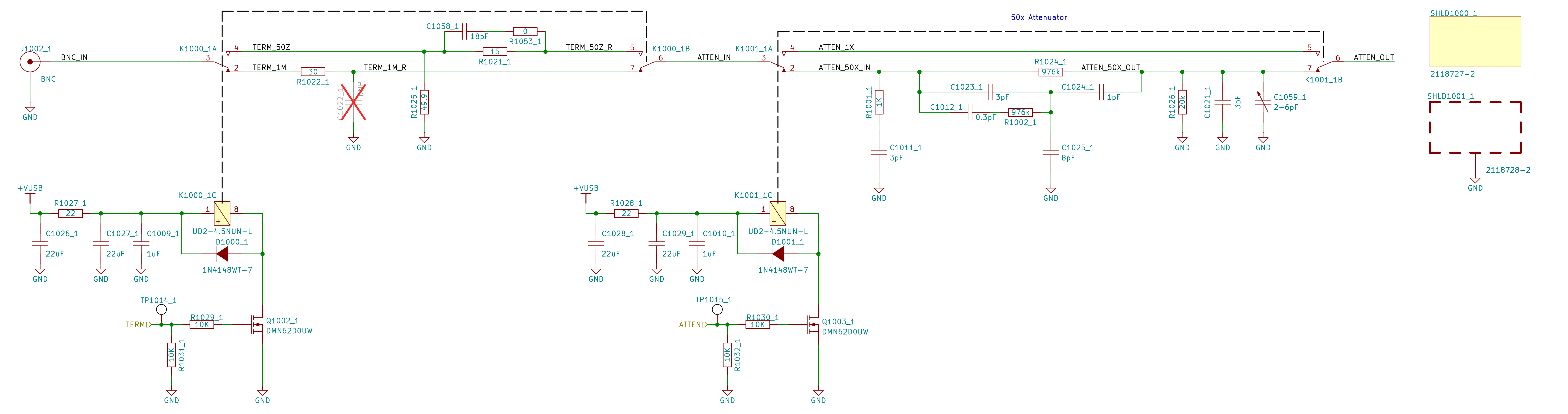


Table 8-3. TMUX4053 Truth Table

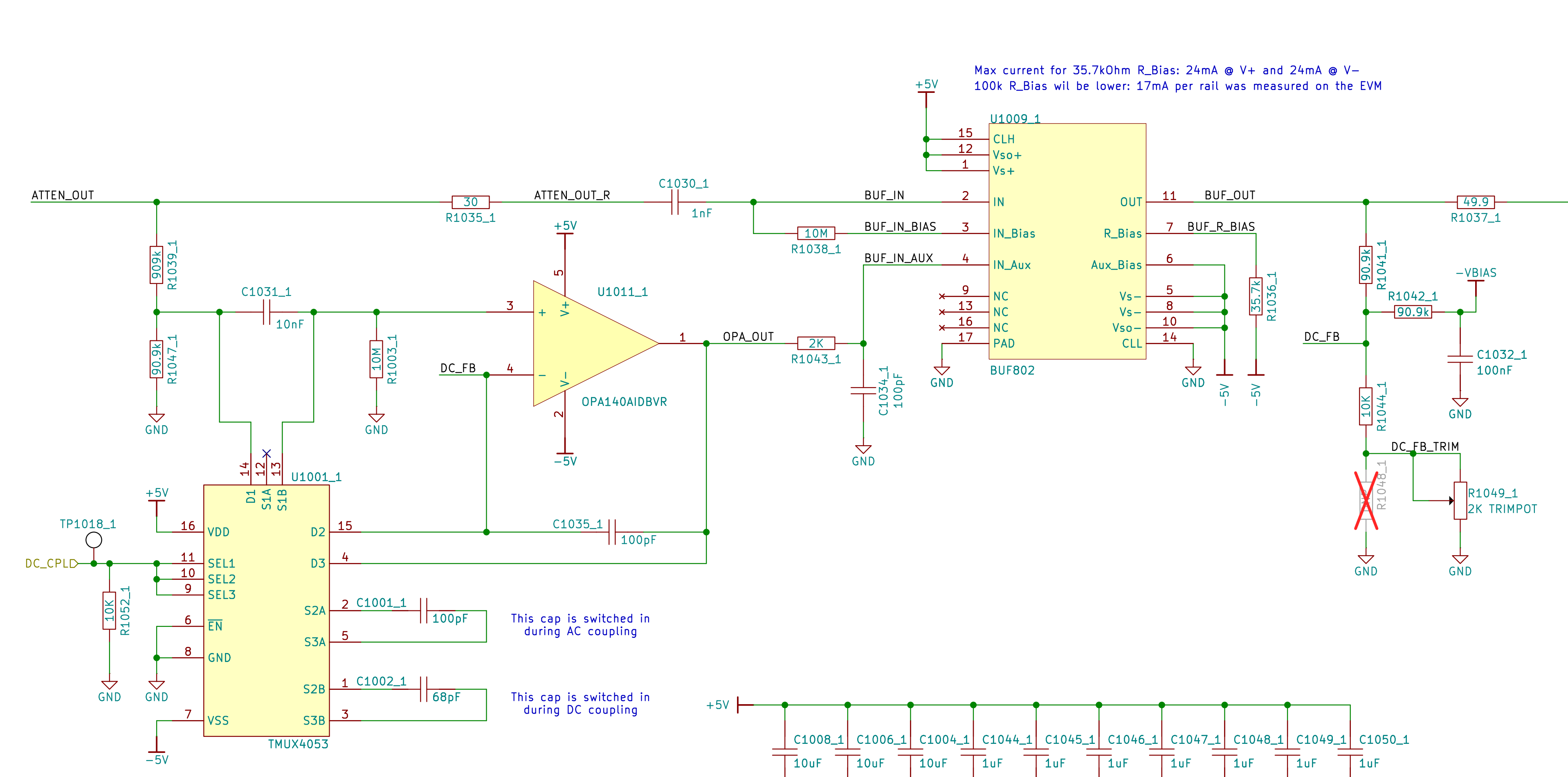
$\overline{EN}$	SEL1	SEL2	SEL3	Selected Signal Path Connected To Drain Pins
0	0	X	X	S1A to D1
0	1	X	X	S1B to D1
0	X	0	X	S2A to D2
0	X	1	X	S2B to D2
0	X	X	0	S3A to D3
0	X	X	1	S3B to D3
1	X(1)	X(1)	X(1)	All inputs are unselected (HI-Z)

EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /CH2/	
File: FE_Channel.kicad_sch	
Title: ThunderScope	
Size: A3 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 2/20

Termination and Attenuation



Input Buffer and AC/DC Coupling



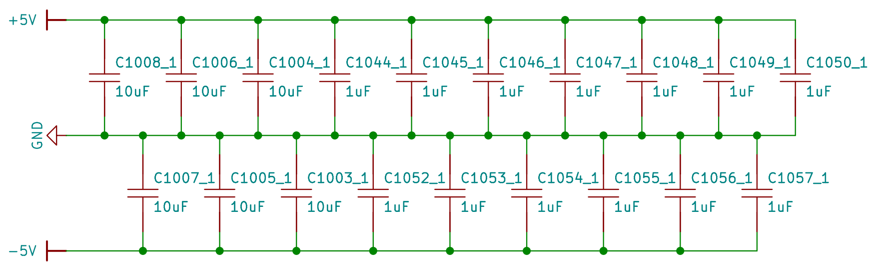
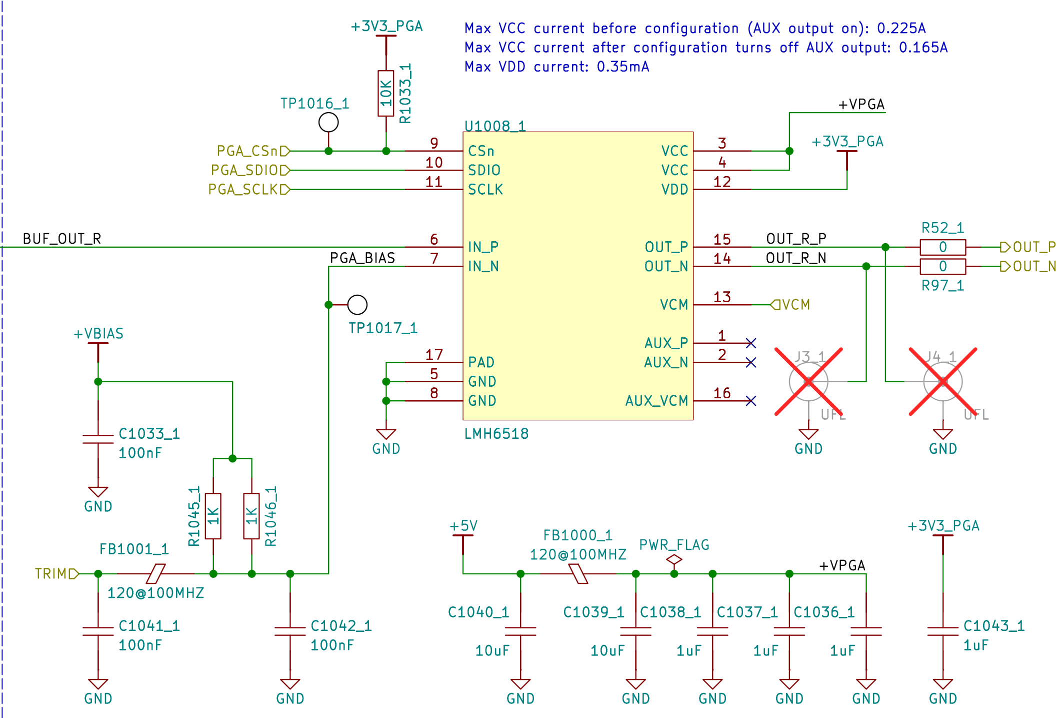


Table 8-3. TMUX4053 Truth Table

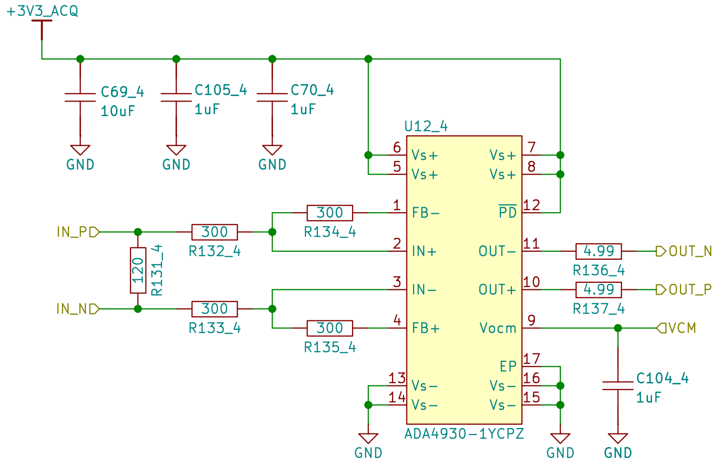
$\overline{EN}$	SEL1	SEL2	SEL3	Selected Signal Path Connected To Drain Pins
0	0	X	X	S1A to D1
0	1	X	X	S1B to D1
0	X	0	X	S2A to D2
0	X	1	X	S2B to D2
0	X	X	0	S3A to D3
0	X	X	1	S3B to D3
1	X[(1)]	X[(1)]	X[(1)]	All inputs are unselected (HI-Z)

Programmable Gain Amplifier



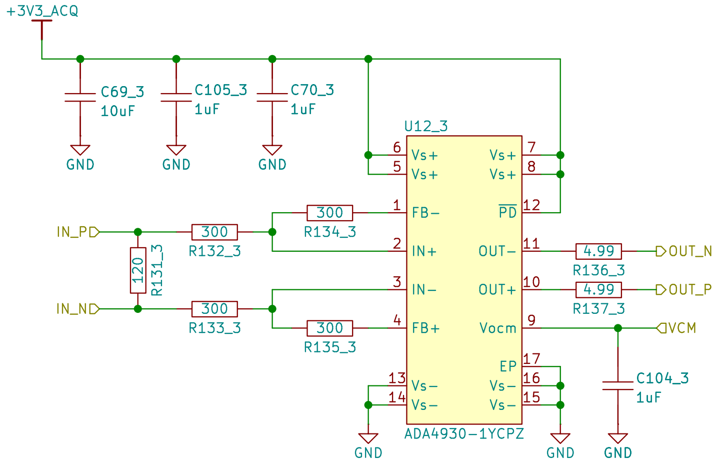
EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /CH1/	
File: FE_Channel.kicad_sch	
Title: ThunderScope	
Size: A3 Date:	Rev: 5.3
KiCad E.D.A. 9.0.3	Id: 2/20

ADC Driver



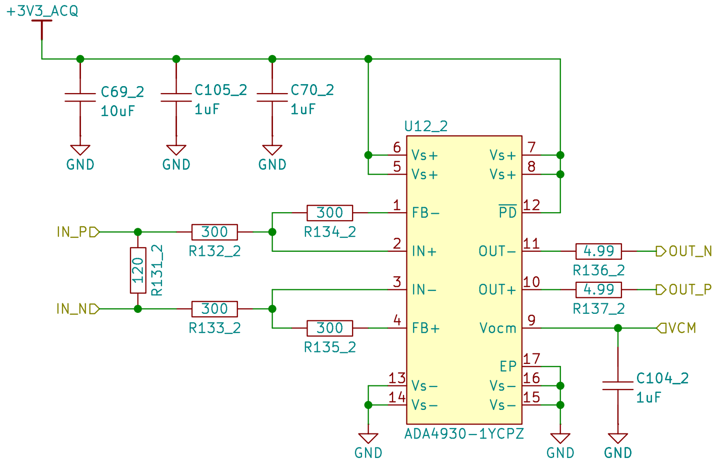
EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /ADC Driver 4/ File: ADC_Driver.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 3/20

ADC Driver



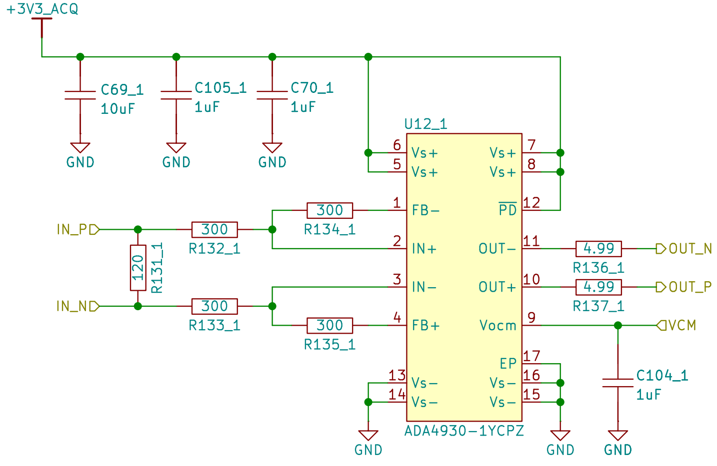
EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /ADC Driver 3/ File: ADC_Driver.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 3/20

ADC Driver



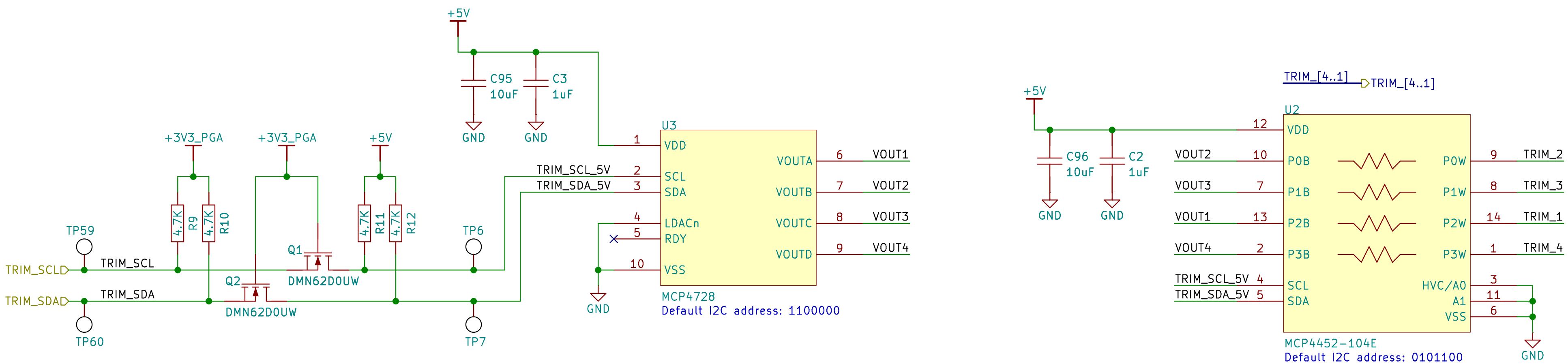
EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /ADC Driver 2/ File: ADC_Driver.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 3/20

ADC Driver

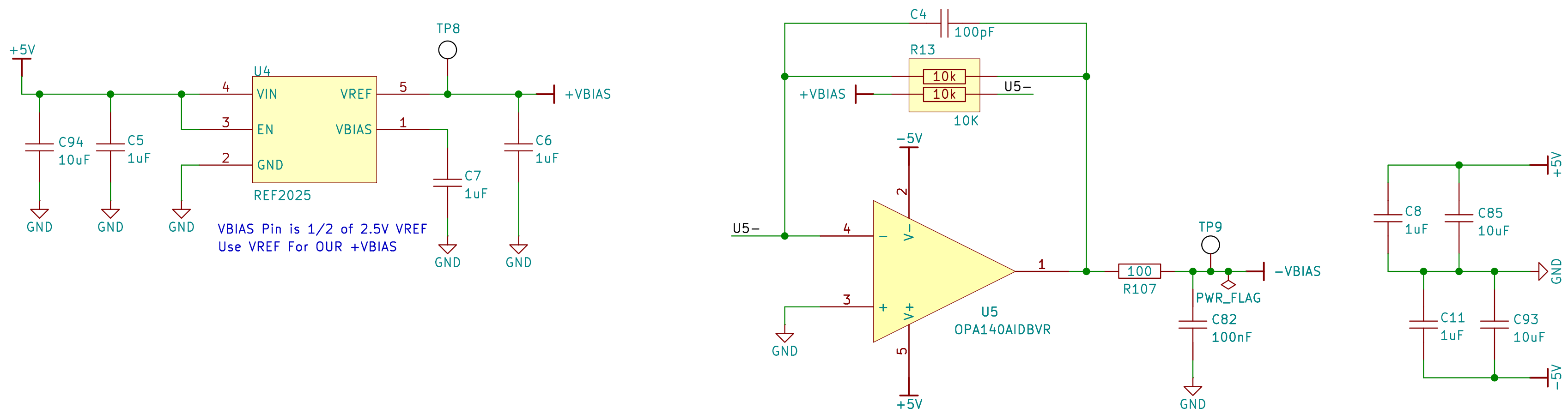


EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /ADC Driver 1/ File: ADC_Driver.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 3/20

Offset Voltage Trim and User Offset Control

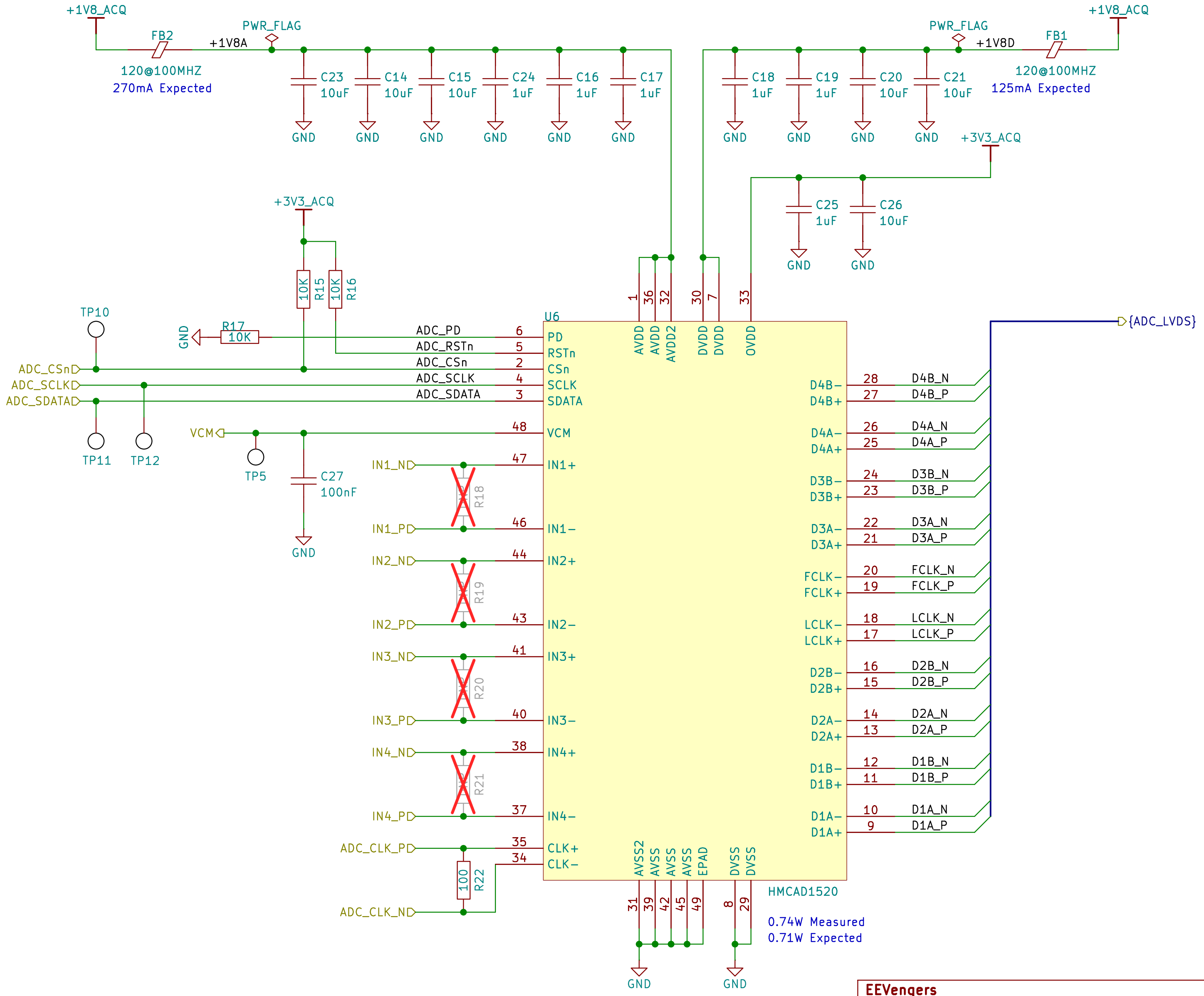


Bias Voltage Generation



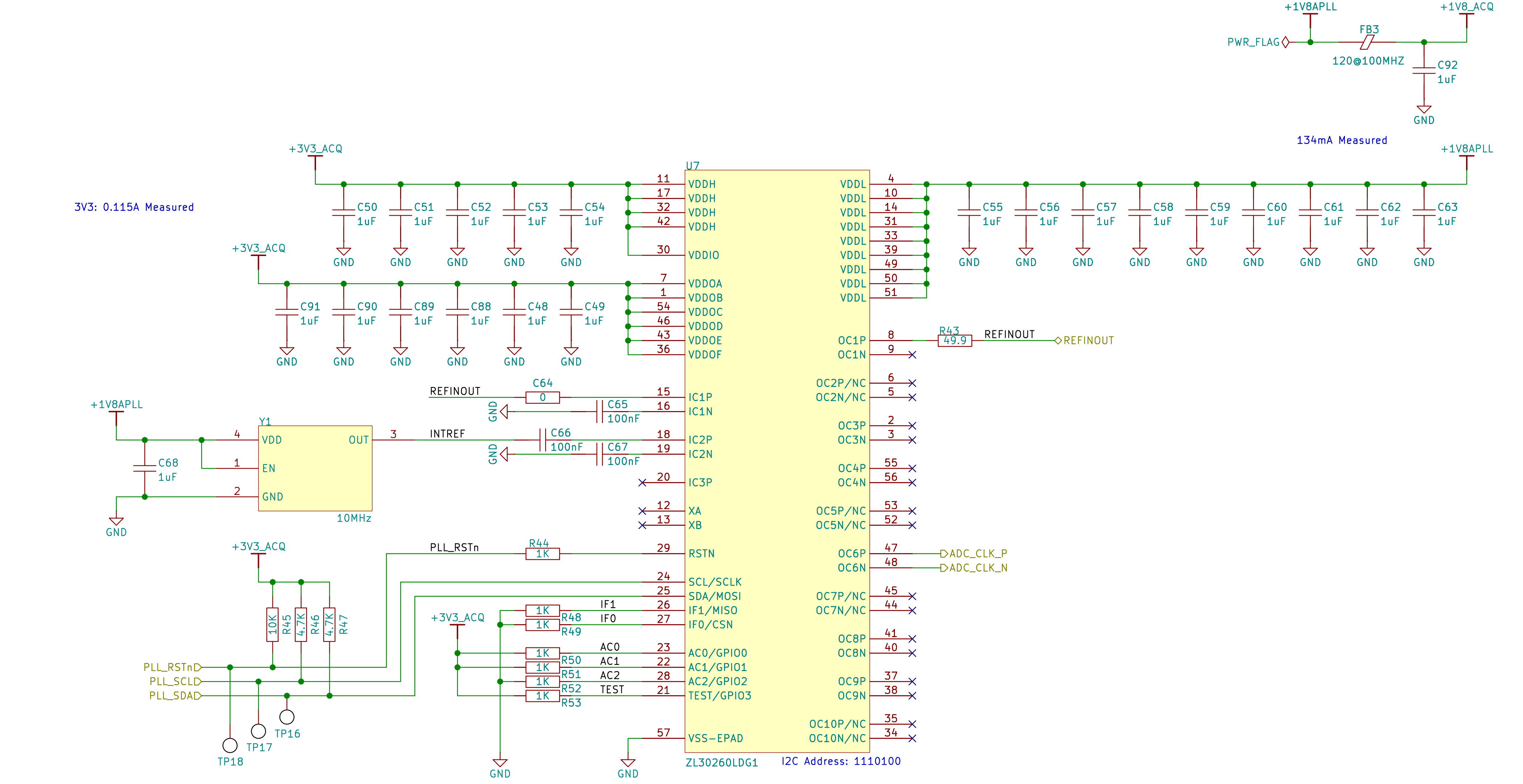
EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /Front End Trim and Bias/ File: FE.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 4/20

ADC



EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /ADC/ File: ADC.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 5/20

Clock Generator



3.3 + 1.8V operation w/ one input one output: 0.67W Expected
0.62W Measured

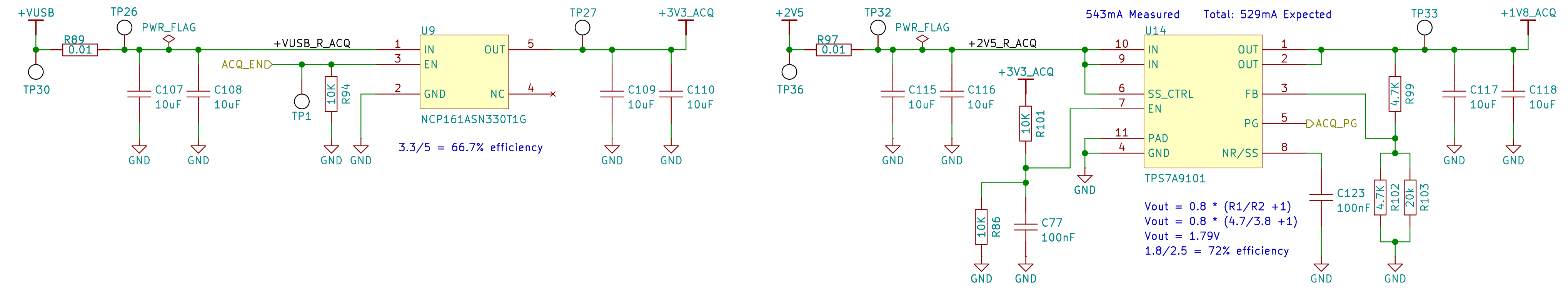
IF1	IF0	Processor Interface	Configuration Memory to Use
0	0	I²C, slave address 11101 00	Internal ROM
0	1	I²C, slave address 11101 01	Internal ROM
1	0	SPI Slave	Internal ROM
1	1	SPI Master during auto-configuration then SPI Slave	External SPI EEPROM

To configure the device as specified in the first three rows above but *without* auto-configuring from internal ROM, wire devices pins as follows: TEST=1 and AC[2:0]=011, as described in section 5.2.

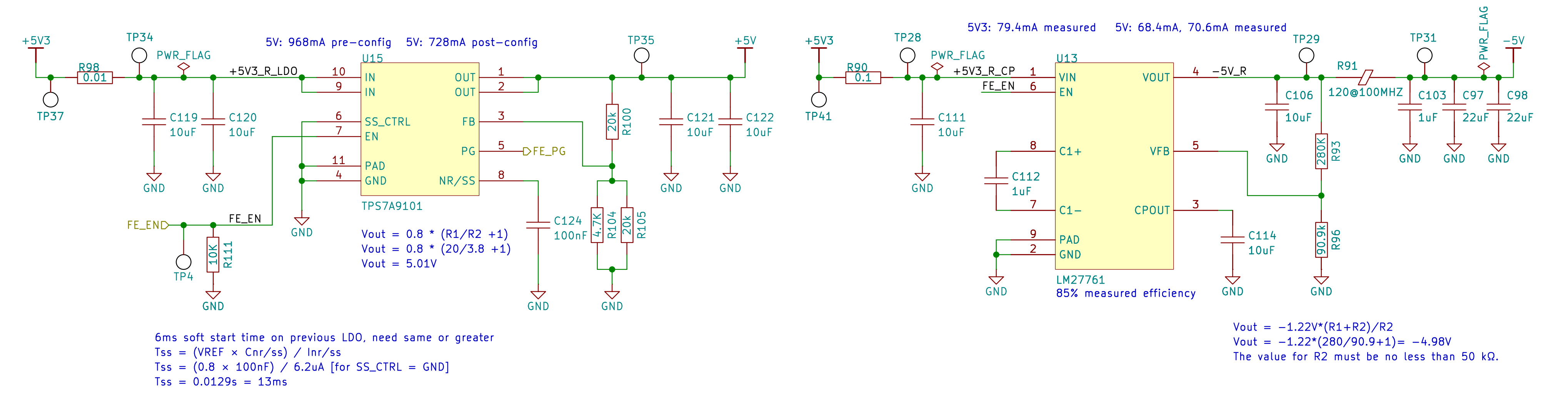
AC2	AC1	AC0	Auto Configuration
0	0	0	Configuration 0
0	0	1	Configuration 1
0	1	0	Configuration 2
0	1	1	Configuration 3
1	0	0	Configuration 4
1	0	1	Configuration 5
1	1	0	Configuration 6
1	1	1	Configuration 7

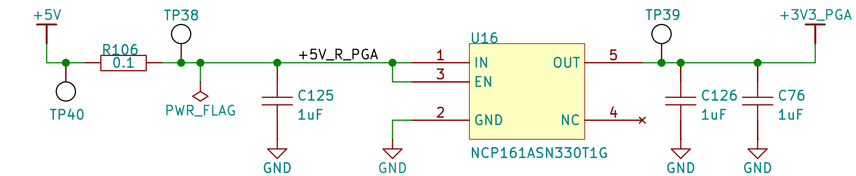
EEVengers Drawn by: Aleksa Bjelogrlic
Sheet: /Clock Generator/
File: PLL.kicad_sch
Title: ThunderScope
Size: A4 Date: **Rev: 5.3**
KiCad E.D.A. 9.0.3 Id: 6/20

Acquisition Voltage Regulators

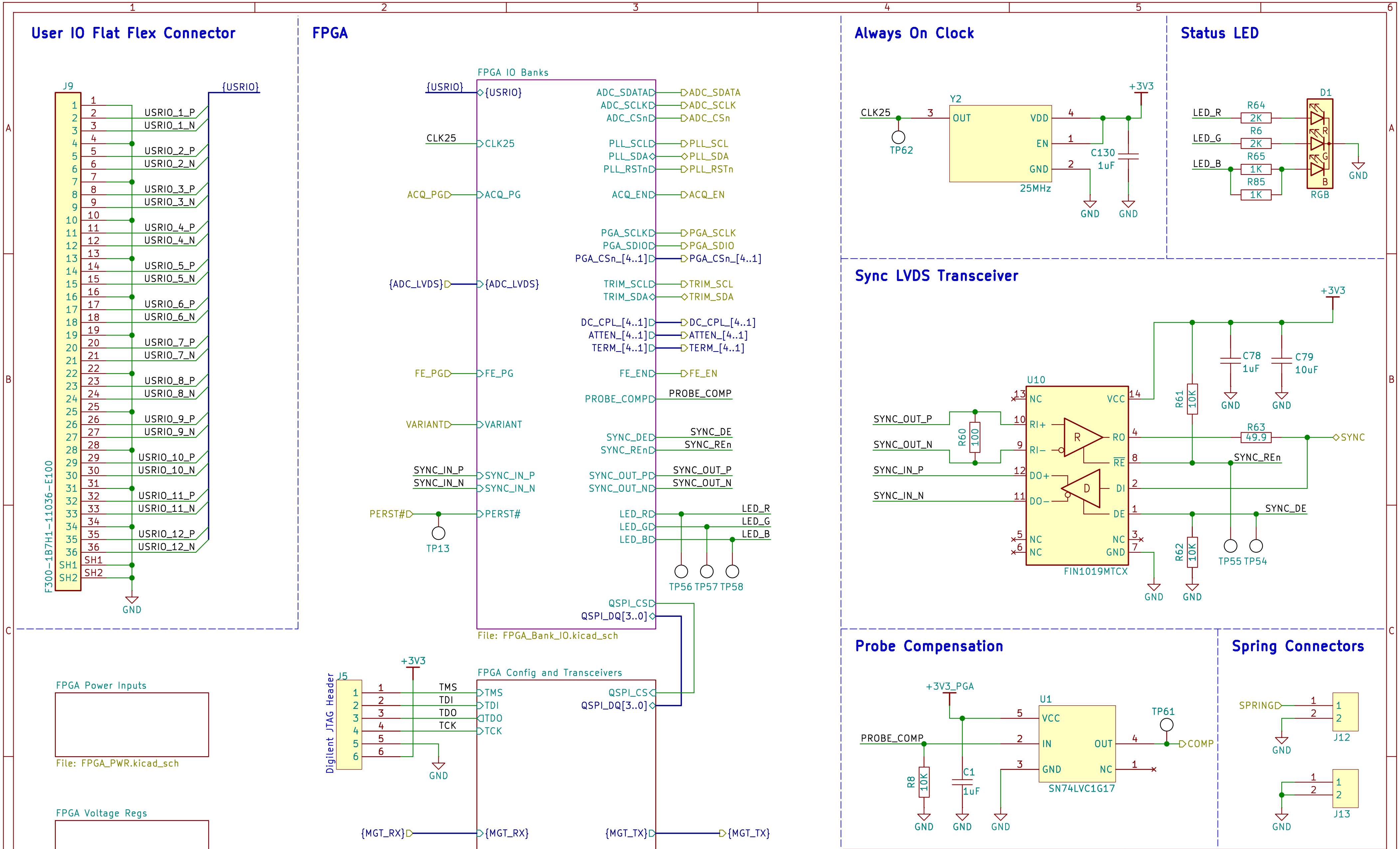


Front End Voltage Regulators





EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /ACQ and FE Voltage Regs/	
File: ACQ_FE_VREG.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 7/20

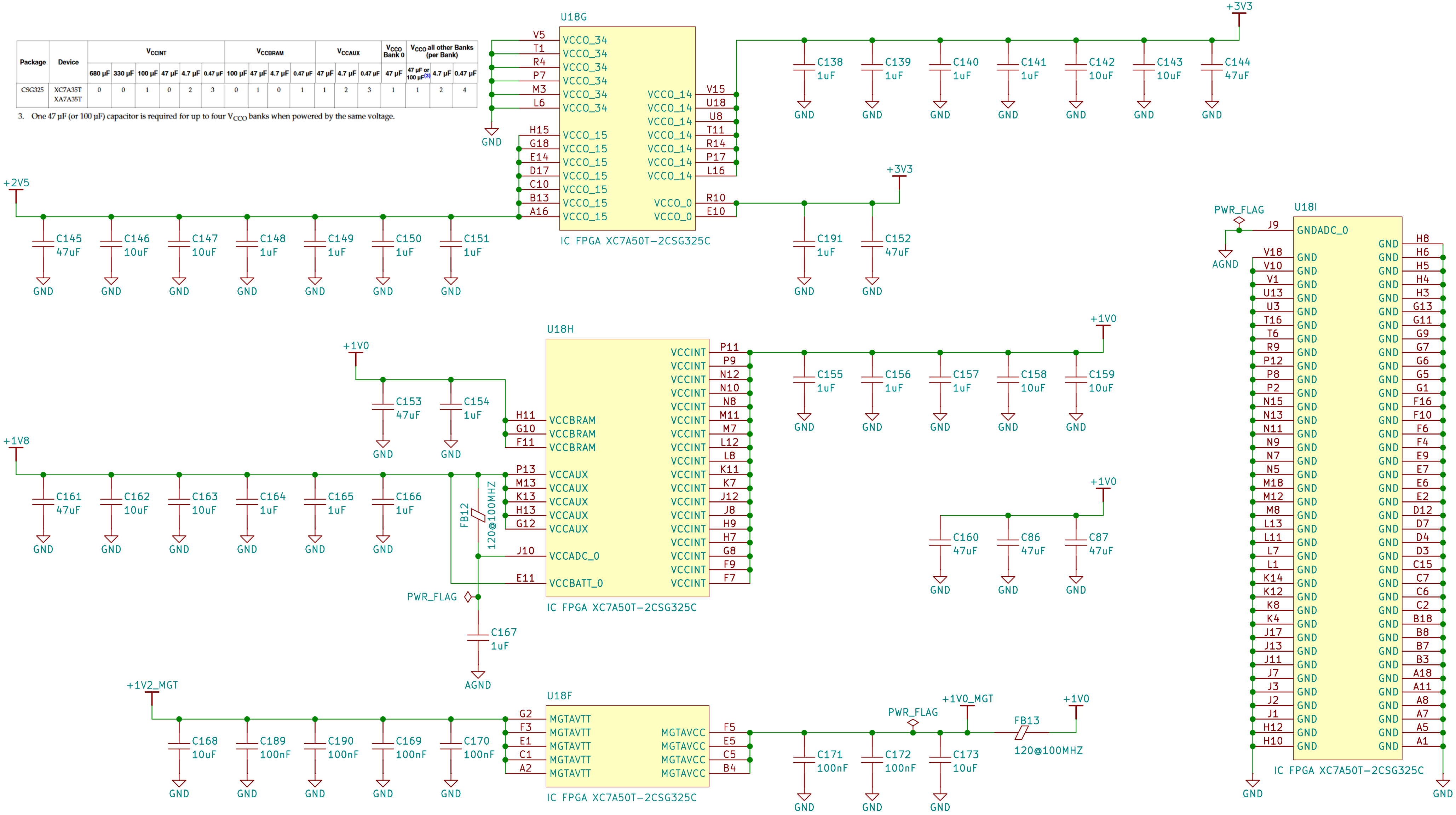


EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /FPGA/	
File: FPGA.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 8/20

FPGA Power Inputs

Package	Device	V_{CCINT}						V_{CCBRAM}				V_{CCAUX}			V_{CCO} Bank 0	V_{CCO} all other Banks (per Bank)		
		680 µF	330 µF	100 µF	47 µF	4.7 µF	0.47 µF	100 µF	47 µF	4.7 µF	0.47 µF	47 µF	4.7 µF	0.47 µF	47 µF	47 µF or 100 µF[3]	4.7 µF	0.47 µF
CSG325	XC7A35T XA7A35T	0	0	1	0	2	3	0	1	0	1	1	2	3	1	1	2	4

3. One 47 µF (or 100 µF) capacitor is required for up to four V_{CCO} banks when powered by the same voltage.



Capacitor	Package Pins			Value
	MGTAVCC	MGTAVTT	GND	
Cap1		F3	F4	0.1 µF
Cap2		A2	A1	
Cap3	B4		B3	
Cap4	F5		F6	

Qty/Power Supply Group		Capacitance (µF)	Tolerance	Type
MGTAVCC	MGTAVTT			
1	1	4.7	10%	Ceramic
2	2	0.1	10%	Ceramic

EEVengers | Drawn by: Aleksa Bjelogrlic

Sheet: /FPGA/FPGA Power Inputs/

File: FPGA_PWR.kicad_sch

Title: ThunderScope

Size: A4 | Date: | **Rev: 5.3**

KiCad E.D.A. 9.0.3 | Id: 11/20

FPGA Voltage Regulators

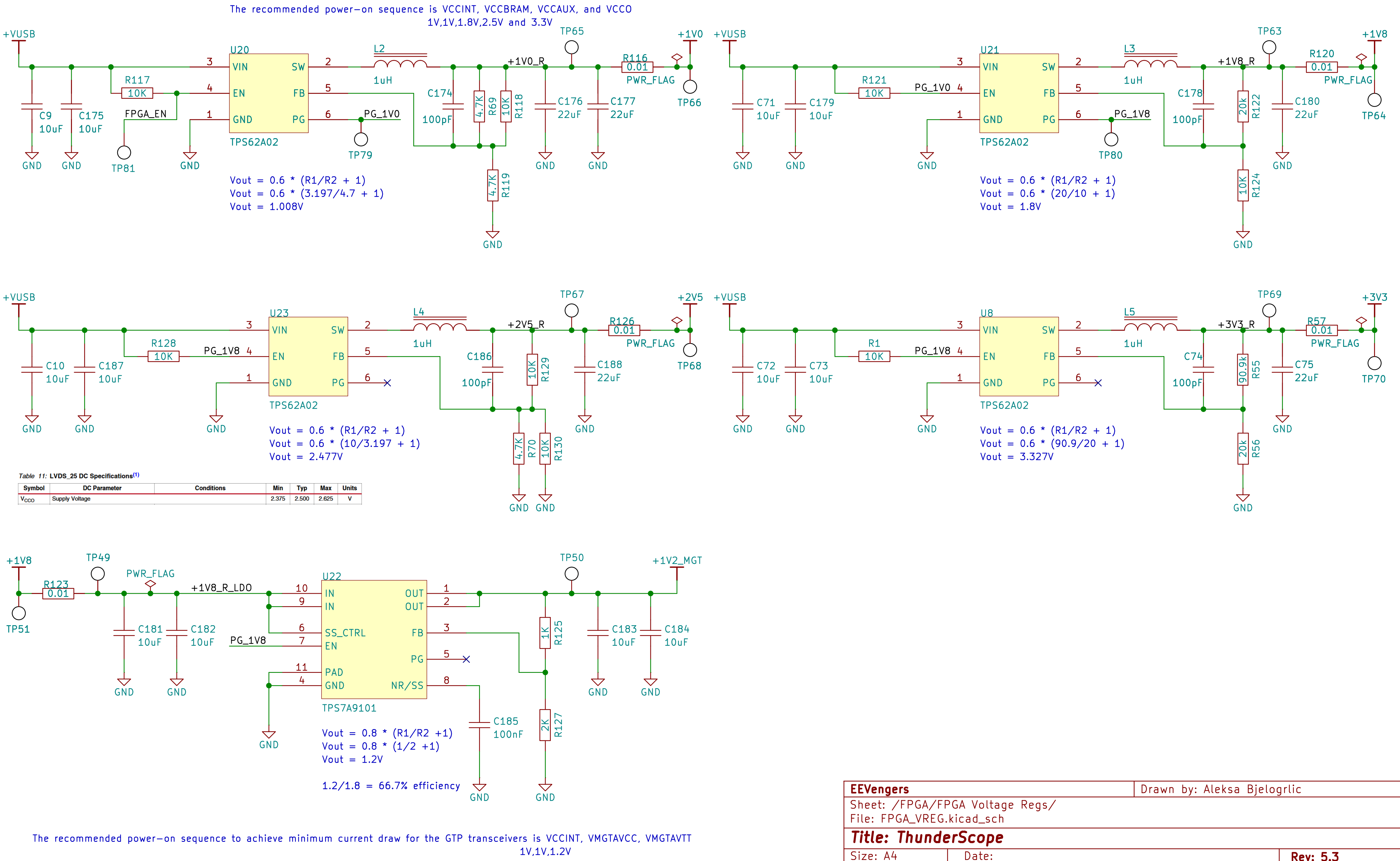
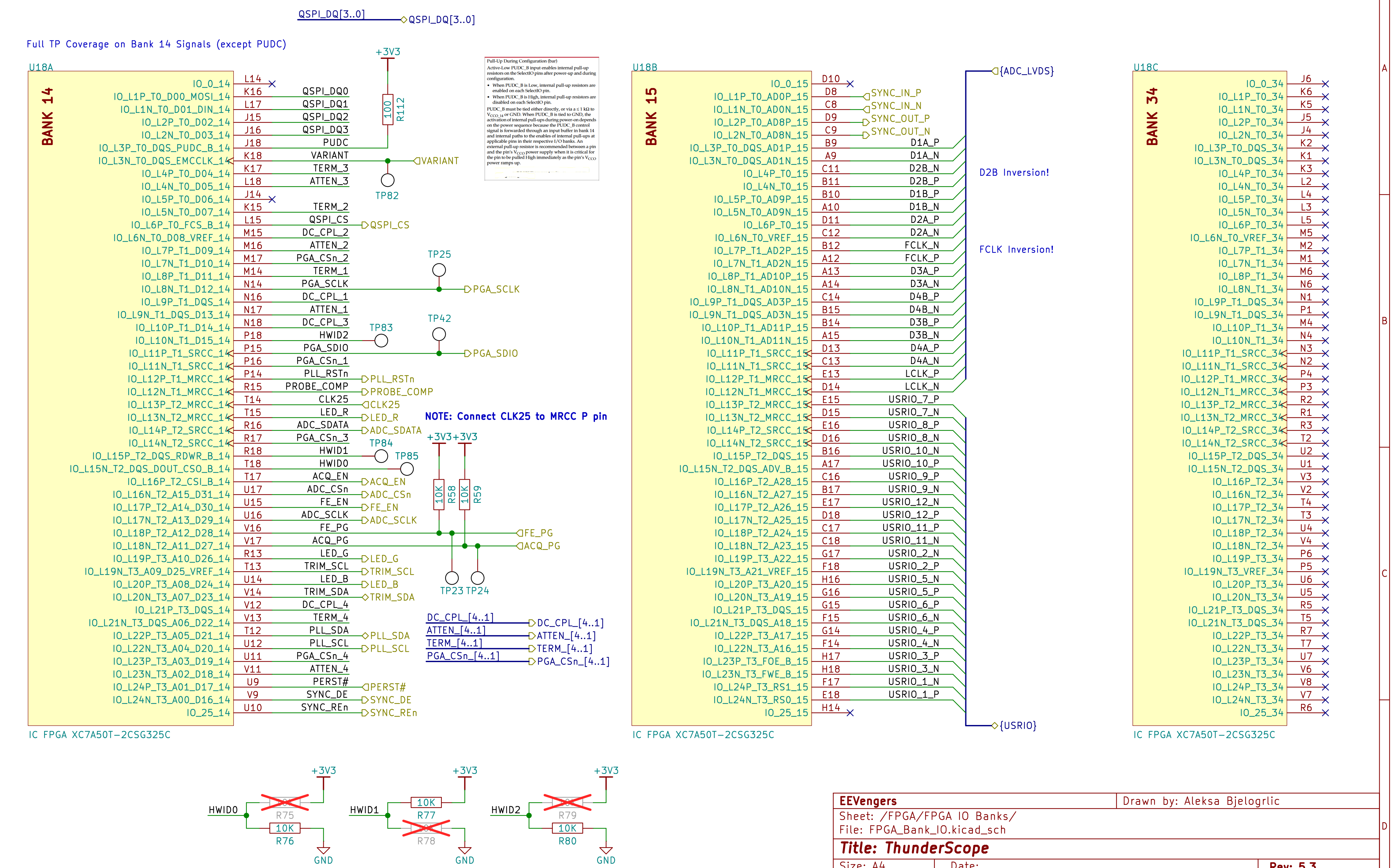


Table 11: **LVDS_25 DC Specifications**(1)

Symbol	DC Parameter	Conditions	Min	Typ	Max	Units
V_{CCO}	Supply Voltage		2.375	2.500	2.625	V

EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /FPGA/FPGA Voltage Regs/	
File: FPGA_VREG.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 12/20

FPGA IO Banks



EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /FPGA/FPGA IO Banks/	
File: FPGA_Bank_IO.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 9/20

FPGA Configuration

A High signal on the DONE pin indicates completion of the configuration sequence. The DONE output is an open-drain output by default.

Note: DONE has an internal pull-up resistor of approximately 10 kΩ. There is no setup/hold requirement for the DONE register. These changes, along with the DonePipe register software default, eliminate the need for the DriveDONE driver-option. External 330Ω resistor circuits are not required but can be used as they have been in previous generations.

Connect INIT_B to a ≤ 4.7 kΩ pull-up resistor to V_{CCO_0} to ensure clean Low-to-High transitions.

Connect PROGRAM_B to an external ≤ 4.7 kΩ pull-up resistor to V_{CCO_0} to ensure a stable High input, and

Table 2-1: **7 Series FPGA Configuration Modes**

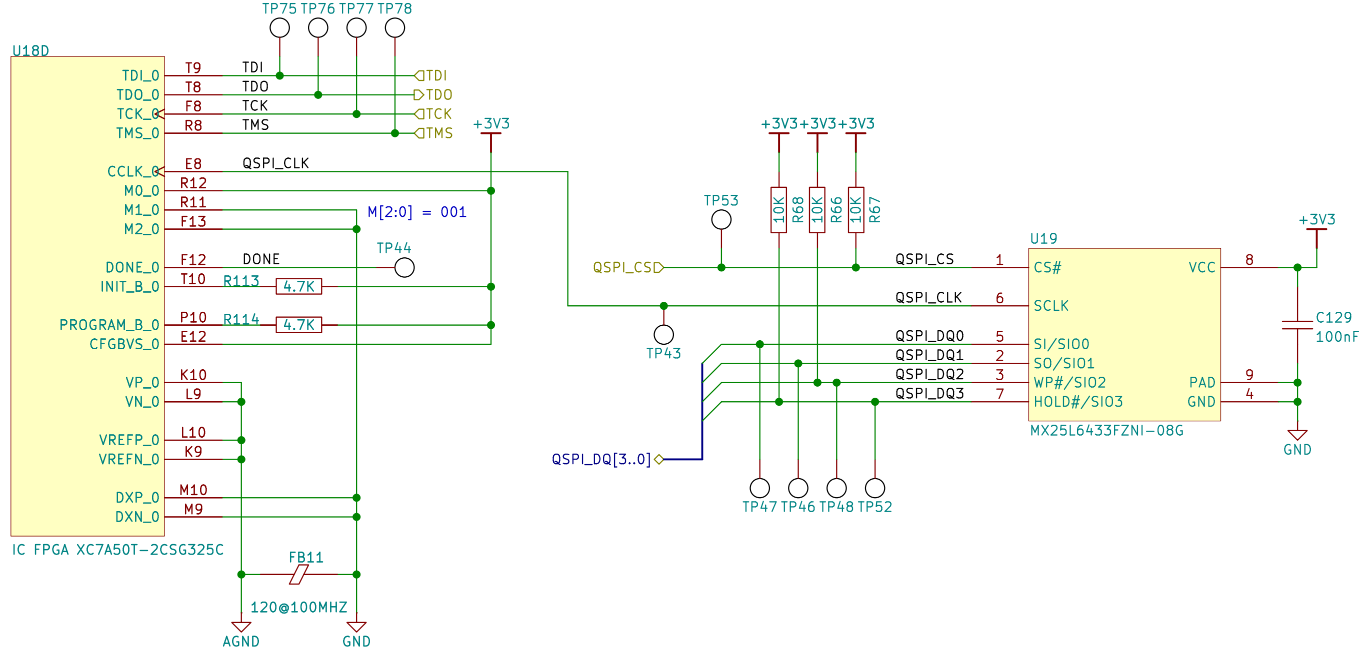
Configuration Mode	M[2:0]	Bus Width	CCLK Direction
Master SPI	001	x1, x2, x4	Output

Table 2-6: **Spartan-7, Artix-7 and Kintex-7 FPGA Configuration Mode, Compatible Voltages, and CFGBVS Connection**

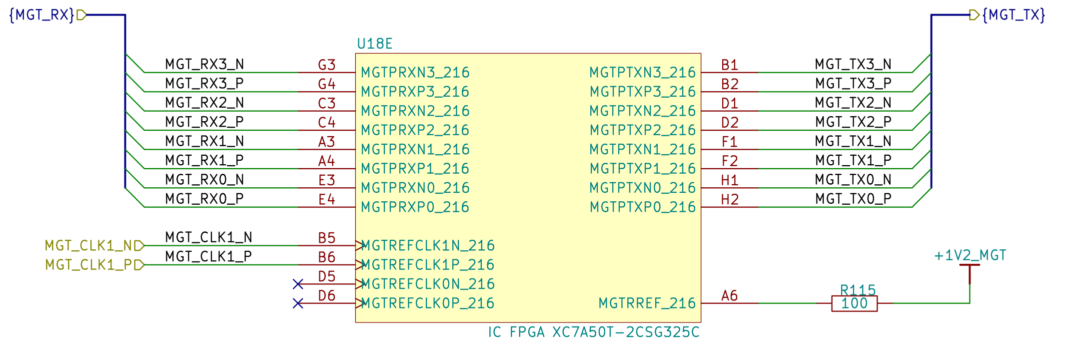
Configuration Mode	Banks Used	Configuration Interface I/O Voltage	HR Bank 0 V_{CCO_0}	HR Bank 14 V_{CCO_14}	HR Bank 15 V_{CCO_15}	CFGBVS
JTAG (only)	0	3.3V	3.3V	Any	Any	VCCO_0
		2.5V	2.5V	Any	Any	VCCO_0
		1.8V	1.8V	Any	Any	GND
		1.5V	1.5V	Any	Any	GND
Serial, SPI, or SelectMAP	0, 14(1)	3.3V	3.3V	3.3V	Any	VCCO_0
		2.5V	2.5V	2.5V	Any	VCCO_0
		1.8V	1.8V	1.8V	Any	GND
		1.5V	1.5V	1.5V	Any	GND
BPI(2)	0, 14, 15	3.3V	3.3V	3.3V	3.3V	VCCO_0
		2.5V	2.5V	2.5V	2.5V	VCCO_0
		1.8V	1.8V	1.8V	1.8V	GND
		1.5V	1.5V	1.5V	1.5V	GND

Notes:

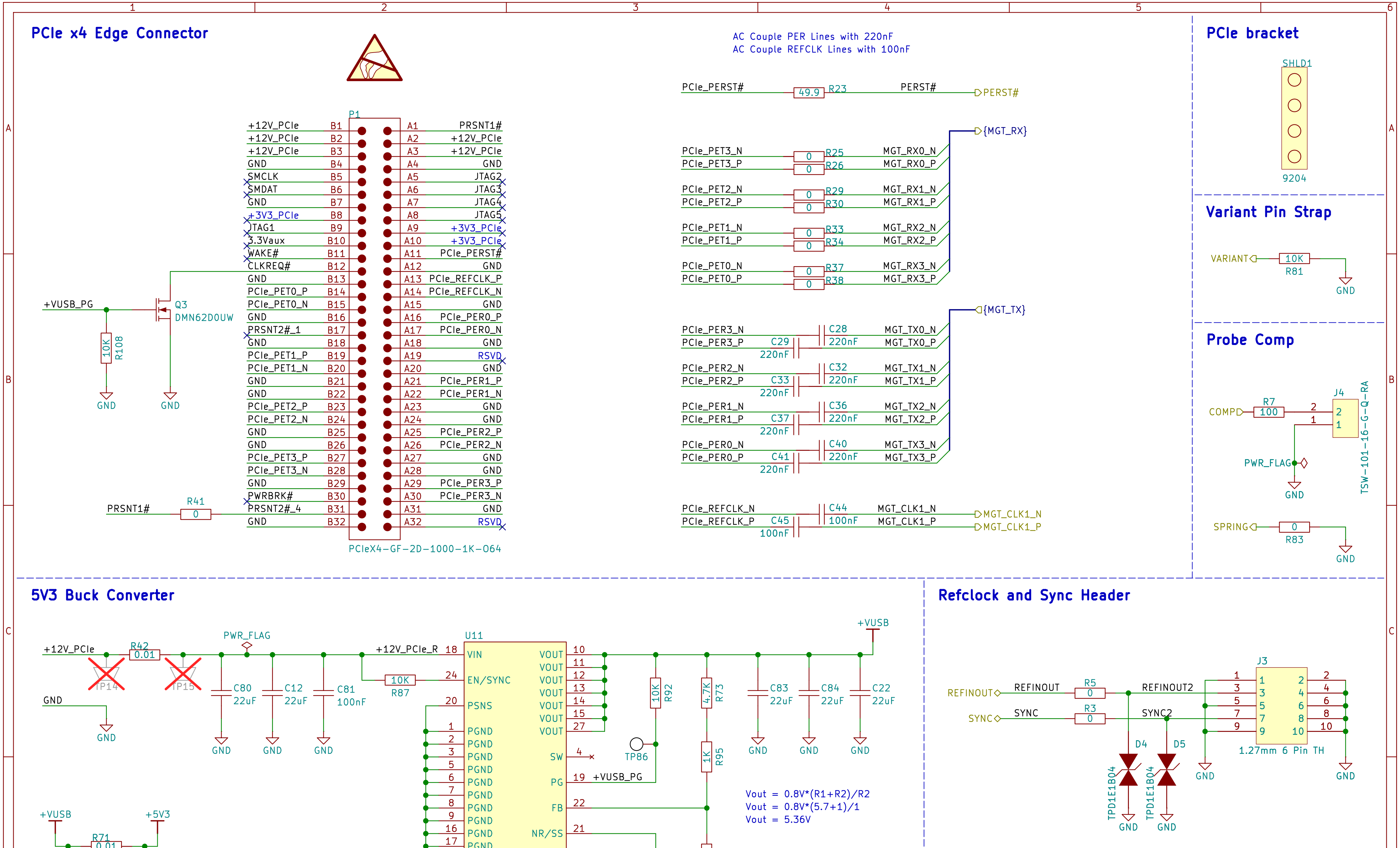
1. RS[1:0] for MultiBoot or Fallback are in bank 15 but are typically only used in BPI mode and not supported in SPI mode.
2. BPI mode is not available in the Spartan-7 family.



FPGA Transceivers



EEVengers	Drawn by: Aleksa Bjelogrlic
Sheet: /FPGA/FPGA Config and Transceivers/ File: FPGA_CFG_MGT.kicad_sch	
Title: ThunderScope	
Size: A4 Date:	**Rev: 5.3**
KiCad E.D.A. 9.0.3	Id: 10/20



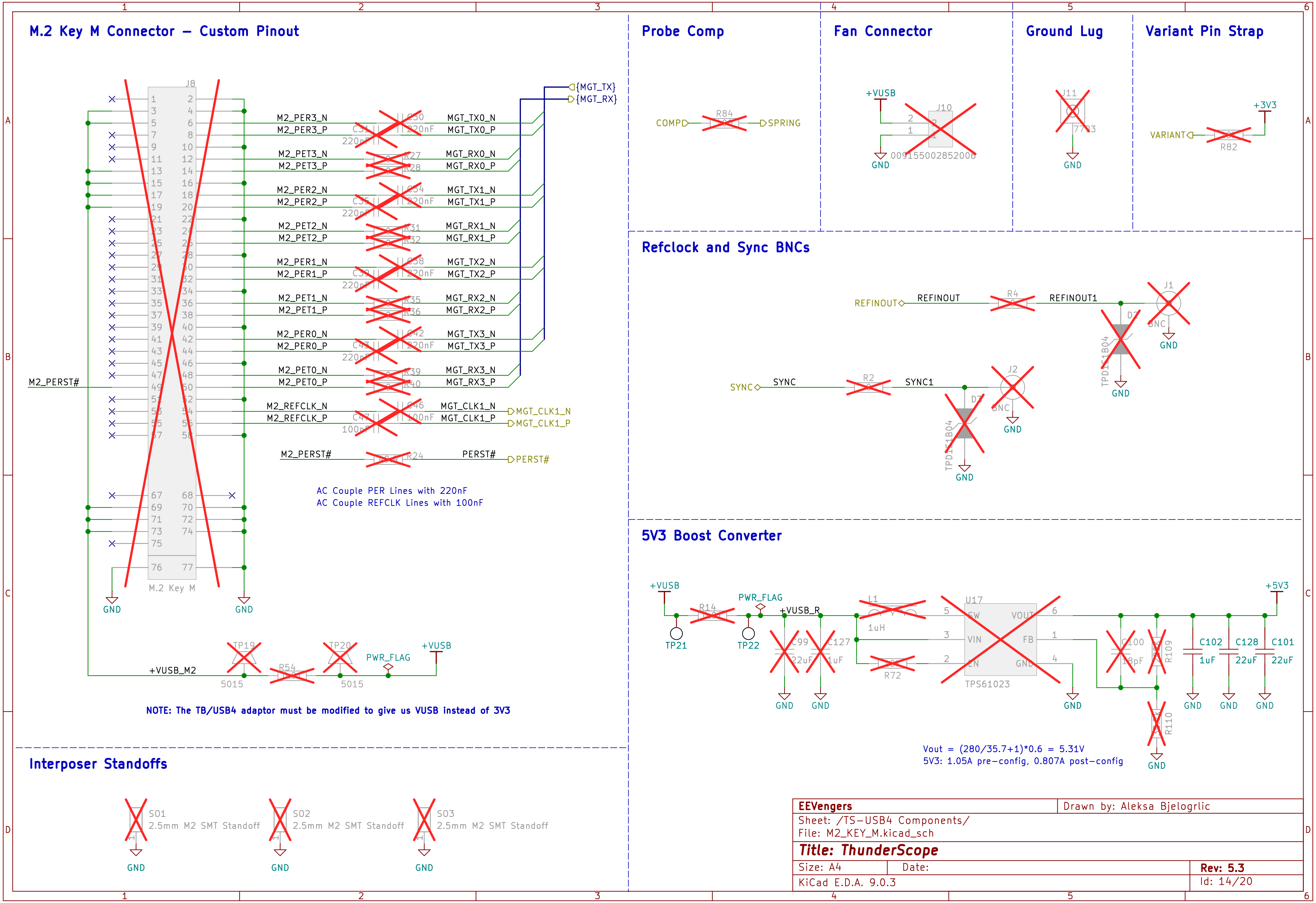
EEVengers | Drawn by: Aleksa Bjelogrlic

Sheet: /TS-PCIe Components/
File: CON_PCIe_X4.kicad_sch

Title: ThunderScope

Size: A4 | Date: | Rev: 5.3

KiCad E.D.A. 9.0.3 | Id: 13/20


M.2 Key M Connector – Custom Pinout
J8
M.2 Key M
M2_PERST#
M2_PER3_N
M2_PER3_P
M2_PET3_N
M2_PET3_P
M2_PER2_N
M2_PER2_P
M2_PET2_N
M2_PET2_P
M2_PER1_N
M2_PER1_P
M2_PET1_N
M2_PET1_P
M2_PER0_N
M2_PER0_P
M2_PET0_N
M2_PET0_P
M2_REFCLK_N
M2_REFCLK_P
MGT_TX0_N
MGT_TX0_P
MGT_RX0_N
MGT_RX0_P
MGT_TX1_N
MGT_TX1_P
MGT_RX1_N
MGT_RX1_P
MGT_TX2_N
MGT_TX2_P
MGT_RX2_N
MGT_RX2_P
MGT_TX3_N
MGT_TX3_P
MGT_RX3_N
MGT_RX3_P
MGT_CLK1_N
MGT_CLK1_P
{MGT_TX}
{MGT_RX}
PERST#
AC Couple PER Lines with 220nF
AC Couple REFCLK Lines with 100nF
+VUSB_M2
PWR_FLAG
+VUSB
GND
NOTE: The TB/USB4 adaptor must be modified to give us VUSB instead of 3V3
Interposer Standoffs
SO1
2.5mm M2 SMT Standoff
SO2
2.5mm M2 SMT Standoff
SO3
2.5mm M2 SMT Standoff
Probe Comp
COMP
R84
SPRING
Fan Connector
+VUSB
J10
009155002852006
Ground Lug
J11
7733
Variant Pin Strap
+3V3
VARIANT
R82
Refclock and Sync BNCs
REFINOUT
R4
REFINOUT1
J1
BNC
TPD1E1B04
SYNC
R2
SYNC1
J2
BNC
5V3 Boost Converter
+VUSB
R14
PWR_FLAG
+VUSB_R
TP21
TP22
L1
1uH
U17
SW
VOUT
VIN
FB
EN
GND
TPS61023
C99
22uF
C127
1uF
R72
C100
18pF
R109
R110
C102
1uF
C128
22uF
C101
22uF
+5V3
Vout = (280/35.7+1)*0.6 = 5.31V
5V3: 1.05A pre-config, 0.807A post-config
EEVengers
Drawn by: Aleksa Bjelogrlic
Sheet: /TS-USB4 Components/
File: M2_KEY_M.kicad_sch
Title: ThunderScope
Size: A4
Date:
Rev: 5.3
KiCad E.D.A. 9.0.3
Id: 14/20